

Tutorial Problems: 1st and 2nd order PDEs

[For tutorial classes in the week commencing 5th February 2024]

1st Order PDEs

1. Find the general solution to the partial differential equation

$$x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} = 0$$

$$P = x, Q = -y, R = 0$$

$$\therefore \frac{dx}{x} = \frac{dy}{-y}$$

$$\therefore \ln y = -\ln x + \ln C_1 = \ln\left(\frac{C_1}{x}\right)$$

$$\therefore y = \frac{C_1}{x} \quad \therefore C_1 = xy$$

$$\therefore du = 0$$

$$\therefore u = C_2 = f(C_1)$$

$$\therefore u = f(xy)$$

2. Find the general solution to the following PDEs and sketch their characteristics

(i)

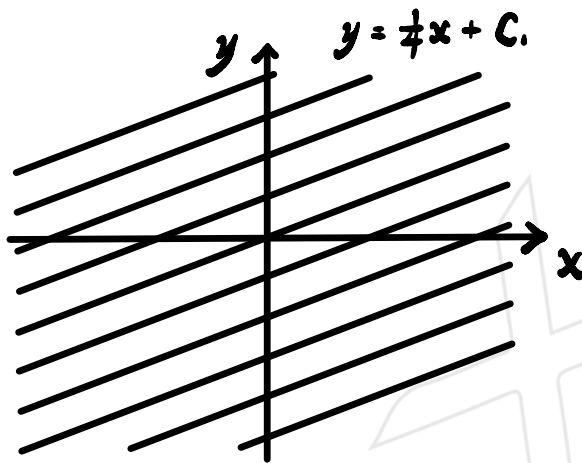
$$4\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2$$

(ii)

$$\frac{\partial u}{\partial x} + (x+y)\frac{\partial u}{\partial y} = 0.$$

i) $P = 4, Q = 1, R = 2$

$$\therefore \frac{dx}{4} = dy \quad \therefore y = \frac{1}{4}x + C_1 \quad \therefore C_1 = y - \frac{1}{4}x$$



$$\therefore \frac{dx}{4} = \frac{du}{2}$$

$$\therefore \frac{1}{2}u = \frac{1}{4}x + \frac{1}{2}C_2$$

$$\therefore u = \frac{1}{2}x + C_2 = \frac{1}{2}x + f(C_1)$$

$$\therefore u = \frac{1}{2}x + f(y - \frac{1}{4}x)$$

2. Find the general solution to the following PDEs and sketch their characteristics

(i)

$$4 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2$$

(ii)

$$\frac{\partial u}{\partial x} + (x+y) \frac{\partial u}{\partial y} = 0.$$

ii) $P = 1, Q = x+y, R = 0$

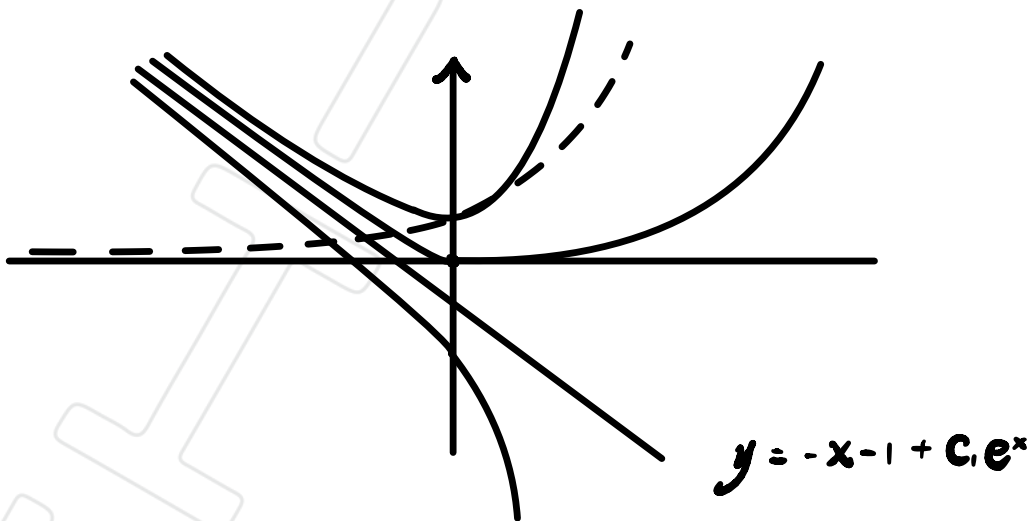
$$\therefore \frac{dx}{1} = \frac{dy}{x+y} \quad \therefore \frac{dy}{dx} = x+y \quad \therefore \frac{dy}{dx} - y = x$$

Let $I = e^{\int -1 dx} = e^{-x} \quad \therefore e^{-x} \frac{dy}{dx} - ye^{-x} = xe^{-x}$

$$\therefore \frac{dIy}{dx} = xe^{-x} \quad \therefore \int dIy = \int xe^{-x} dx$$

$$\therefore e^{-x} y = -xe^{-x} - e^{-x} + C_1 \quad \therefore y = -x - 1 + C_1 e^x$$

0	1
+	$x e^x$
-	$1 \cdot e^{-x}$
+	$0 \cdot e^x$



$$du = 0 \quad \therefore u = C_2 = f(C_1) = f\left(\frac{x+y+1}{e^x}\right)$$

3. Find the general solution to the PDE

$$\frac{1}{x} \frac{\partial u}{\partial x} + (x^2 - y) \frac{\partial u}{\partial y} = \frac{1}{x + x^3}$$

$$P = \frac{1}{x}, \quad Q = x^2 - y, \quad R = \frac{1}{x + x^3}$$

$$\therefore \frac{dx}{(\frac{1}{x})} = \frac{dy}{x^2 - y} \quad \therefore \frac{dy}{dx} = x^2 - xy$$

$$\therefore \frac{dy}{dx} + xy = x^2$$

$$\text{Let } I = e^{\int x \, dx} = e^{\frac{1}{2}x^2} \quad \therefore e^{\frac{1}{2}x^2} \frac{dy}{dx} + xy e^{\frac{1}{2}x^2} = x^2 e^{\frac{1}{2}x^2}$$

$$\therefore \frac{dIy}{dx} = x^2 e^{\frac{1}{2}x^2} \quad \therefore Iy = \int x^2 e^{\frac{1}{2}x^2} dx$$

$$\text{Let } z = x^2 \quad \therefore \frac{dz}{dx} = 2x \quad \therefore Iy = \int \frac{1}{2} z e^{\frac{1}{2}z} dz$$

$$\therefore Iy = z e^{\frac{1}{2}z} - 2 e^{\frac{1}{2}z} + C_1$$

$$= x^2 e^{\frac{1}{2}x^2} - 2 e^{\frac{1}{2}x^2} + C_1$$

$$\therefore y = x^2 - 2 + C_1 e^{-\frac{1}{2}x^2}$$

$$\therefore C_1 = (y - x^2 + 2) e^{\frac{1}{2}x^2}$$

$$\frac{dx}{(\frac{1}{x})} = \frac{du}{(\frac{1}{x+x^3})} \quad \therefore \int \frac{1}{1+x^2} dx = \int du$$

$$\therefore u = \tan^{-1}(x) + C_2$$

$$= \tan^{-1}(x) + f[(y - x^2 + 2) e^{\frac{1}{2}x^2}]$$

0	1
+	$\frac{1}{2}z$
-	$\frac{1}{2}$
+	0
	$4e^{\frac{1}{2}z}$

4. Show that the partial differential equation

$$y^2 u_x + x^2 u_y = 2xy^2$$

has general solution given by

$$u(x, y) = x^2 + f(x^3 - y^3).$$

If the equation has boundary conditions $u = -y^6$ when $x = 0$ find the particular solution.

$$P = y^2, \quad Q = x^2, \quad R = 2xy^2$$

$$\therefore \frac{dx}{y^2} = \frac{dy}{x^2}$$

$$\therefore \frac{1}{3}y^3 = \frac{1}{3}x^3 + C_1, \quad \therefore C_1 = \frac{1}{3}y^3 - \frac{1}{3}x^3$$

$$\therefore \frac{dx}{y^2} = \frac{du}{2xy^2}$$

$$\therefore \int du = \int 2x \, dx \quad \therefore u = x^2 + C_2 = x^2 + f(C_1)$$

$$\therefore u = x^2 + f\left(\frac{1}{3}y^3 - \frac{1}{3}x^3\right)$$

$$\text{for } x = 0, \quad u = -y^6$$

$$\therefore u = f\left(\frac{1}{3}y^3\right) = -y^6 \quad \therefore f(t) = -(3t)^2$$

$$\therefore u = x^2 - (y^3 - x^3)^2$$

5. Show that the general solution of the PDE

$$\frac{\partial u}{\partial x} + 2x \frac{\partial u}{\partial y} + 4x = 0$$

is $u(x, y) = -2x^2 + f(x^2 - y)$, where f is an arbitrary function.

If $u = 2$ when $y = \frac{3}{2}x^2$ find the particular solution and state the range of values for which the solution is defined. Comment on the uniqueness of the solution.

$$P = 1, \quad Q = 2x, \quad R = -4x$$

$$\therefore \frac{dx}{1} = \frac{dy}{2x}$$

$$\therefore \int dy = \int 2x dx \quad \therefore y = x^2 + C_1$$

$$\therefore C_1 = y - x^2$$

$$\therefore \frac{dx}{1} = \frac{du}{-4x}$$

$$\therefore \int du = \int -4x dx$$

$$\therefore u = -2x^2 + C_2 = -2x^2 + f(C_1)$$

$$\therefore u = -2x^2 + f(y - x^2)$$

$$\text{When } y = \frac{3}{2}x^2, \quad u = 2$$

$$\therefore u = -2x^2 + f\left(\frac{1}{2}x^2\right) = 2$$

$$\therefore -4t + f(t) = 2 \quad \therefore f(t) = 4t + 2$$

$$\therefore u = -2x^2 + 4(y - x^2) + 2 \quad \text{for } y - x^2 \geq 0$$

6. Show that the general solution to the partial differential equation

$$x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} = (x + y)u$$

is given by

$$u(x, y) = (x - y)f\left(\frac{1}{y} - \frac{1}{x}\right)$$

for an arbitrary function f .

[Hint: This is not the only correct version of the solution. If you find a different solution, can you modify it to have this form?]

$$P = x^2, \quad Q = y^2, \quad R = (x + y)u$$

$$\therefore \frac{dx}{x^2} = \frac{dy}{y^2} \quad \therefore \frac{dy}{dx} = \frac{y^2}{x^2}$$

$$\therefore -\frac{1}{y} = -\frac{1}{x} + C_1 \quad \therefore C_1 = -\frac{1}{y} + \frac{1}{x}$$

$$\therefore \frac{dx}{x^2} = \frac{du}{(x + y)u}$$

$$\therefore \frac{du}{u} = \frac{x + y}{x^2} dx = \frac{1}{x} dx + \frac{y}{x^2} dx = \frac{dx}{x} + \frac{dy}{y}$$

$$\therefore \ln|u| = \ln|x| + \ln|y| + \ln|C_1|$$

$$\therefore \ln|u| = \ln|C_1 xy| = \ln|f(C_1)xy|$$

$$\therefore u = xy f\left(-\frac{1}{y} + \frac{1}{x}\right)$$

$$f\left(-\frac{1}{y} + \frac{1}{x}\right) = \left(\frac{1}{xy} - \frac{1}{xy}\right)g\left(-\frac{1}{y} + \frac{1}{x}\right)$$

$$\therefore u = (x - y)g\left(-\frac{1}{y} + \frac{1}{x}\right) \text{ for } \boxed{xy = x - y}$$

2nd Order PDEs

7. Show that the PDE

$$u_{xx} + 2u_{xy} + u_{yy} = 0$$

is parabolic and find its general solution.

[Hint: the choice $\eta(x, y) = y$ is a valid choice - why?]

$$R = 1, S = -1, T = 1, f = 0$$

$$\therefore S^2 - RT = 1 - 1 = 0 \quad \therefore \text{parabolic}$$

$$\therefore \frac{dy}{dx} = \frac{-S \pm \sqrt{S^2 - RT}}{R} = 1 \quad \therefore y = x + C_1$$

$$\therefore \xi(x, y) = C_1 = y - x$$

$$\therefore \xi_x = -1, \xi_y = 1$$

$$\therefore \xi_x \eta_y - \xi_y \eta_x \neq 0$$

$$\therefore -\eta_y - \eta_x \neq 0 \quad \therefore \eta(x, y) = y$$

$$\therefore \eta_x = 0, \eta_y = 1$$

$$\therefore u_x = u_\xi \xi_x + u_\eta \eta_x = -u_\xi$$

$$u_y = u_\xi \xi_y + u_\eta \eta_y = u_\xi + u_\eta$$

$$\therefore u_{xx} = \frac{\partial}{\partial x} u_x = \left(\frac{\partial}{\partial \xi} \xi_x + \frac{\partial}{\partial \eta} \eta_x \right) (-u_\xi) = u_{\xi\xi}$$

$$u_{yy} = \frac{\partial}{\partial y} u_y = \left(\frac{\partial}{\partial \xi} \xi_y + \frac{\partial}{\partial \eta} \eta_y \right) (u_\xi + u_\eta) = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}$$

$$u_{xy} = -u_{\xi\xi} - u_{\xi\eta}$$

$$\therefore u_{xx} + 2u_{xy} + u_{yy} = u_{\eta\eta} = 0 \quad \therefore u_{\eta\eta} = 0$$

$$\therefore u_\eta = \int u_{\eta\eta} d\eta = F(\xi)$$

$$\therefore u = \int u_\eta d\eta = \int F(\xi) d\eta = \eta f(\xi) + g(\xi) = y f(y-x) + g(y-x)$$

8. Classify the PDE

$$\frac{\partial^2 u}{\partial x^2} - x^2 \frac{\partial^2 u}{\partial y^2} = 0$$

reduce it to standard form and sketch its characteristics.

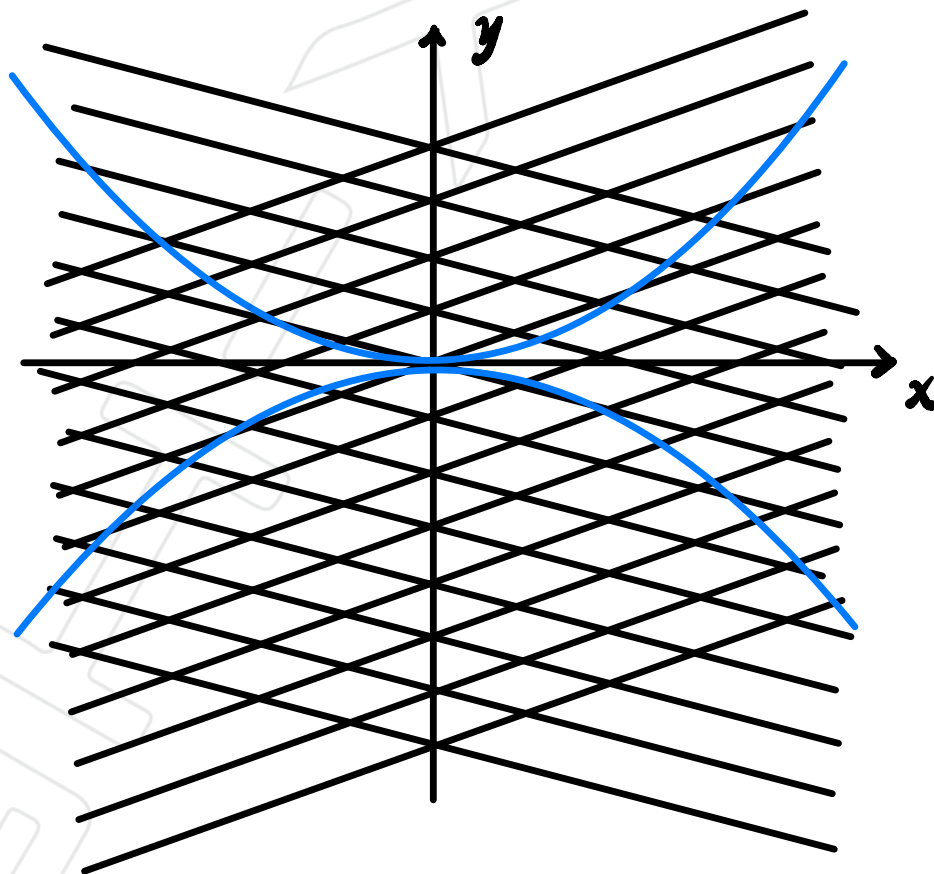
$$R = 1, S = 0, T = -x^2, f = 0 \quad (x \neq 0)$$

$$\therefore S^2 - RT = 0 + x^2 > 0 \quad \therefore \text{hyperbolic}$$

$$\therefore \frac{dy}{dx} = \frac{-S \pm \sqrt{S^2 - RT}}{R} = \pm x$$

$$\therefore \begin{cases} y = \frac{1}{2}x^2 + C_1 \\ y = -\frac{1}{2}x^2 + C_2 \end{cases}$$

$$\therefore \begin{cases} \xi = C_1 = y - \frac{1}{2}x^2 \\ \eta = C_2 = y + \frac{1}{2}x^2 \end{cases}$$



$$\therefore \xi_x = -x, \xi_y = 1, \eta_x = x, \eta_y = 1$$

$$\therefore u_x = u_\xi \xi_x + u_\eta \eta_x = -x u_\xi + x u_\eta$$

$$u_y = u_\xi \xi_y + u_\eta \eta_y = u_\xi + u_\eta$$

$$\therefore u_{xx} = \underbrace{-u_\xi + u_\eta}_{-u_\xi + u_\eta} + x^2 u_{\xi\xi} + x^2 u_{\eta\eta} - 2x^2 u_{\xi\eta}$$

$$u_{yy} = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}$$

$$u_{xy} = x u_{\eta\eta} - x u_{\xi\xi}$$

$$\therefore u_{xx} - x^2 u_{yy} = \underbrace{-u_\xi + u_\eta - 4x^2 u_{\xi\eta}}_{-4x^2 u_{\xi\eta} = 0} = 0$$

$$\therefore \underline{u_{\xi\eta} = 0} \quad u_{\xi\eta} = \frac{u_\eta - u_\xi}{4x^2} = \frac{u_\eta - u_\xi}{4(\eta - \xi)}$$

$$\therefore \underline{u_\xi = \int u_{\xi\eta} d\eta = \int 0 d\eta = F(\xi)}$$

$$\therefore \underline{u = \int u_\xi d\xi = \int F(\xi) d\xi = f(\xi) + g(\eta)}$$

$$\therefore \underline{u = f(y - \frac{1}{2}x^2) + g(y + \frac{1}{2}x^2)}$$

9. Classify the PDE

$$u_{xx} + x^4 u_{yy} = 0$$

and find its standard form.

$$R = 1, \quad S = 0, \quad T = x^4, \quad f = 0 \quad (x \neq 0)$$

$$\therefore S^2 - RT = -x^4 < 0 \quad \therefore \text{Elliptic}$$

$$\therefore \frac{dy}{dx} = \frac{-S \pm \sqrt{S^2 - RT}}{R} = \pm ix^2$$

$$\therefore \begin{cases} y = \frac{1}{3}ix^3 + C_1 \\ y = -\frac{1}{3}ix^3 + C_2 \end{cases}$$

$$\therefore \begin{cases} \xi(x, y) = y \\ \eta(x, y) = \frac{1}{3}x^3 \end{cases}$$

$$\therefore \xi_x = 0, \quad \xi_y = 1, \quad \eta_x = x^2, \quad \eta_y = 0$$

$$\therefore u_x = u_\xi \xi_x + u_\eta \eta_x = x^2 u_\eta$$

$$u_y = u_\xi \xi_y + u_\eta \eta_y = u_\xi$$

$$\therefore u_{xx} = \frac{2x u_\eta + x^4 u_{\eta\eta}}{1}$$

$$u_{yy} = u_{\xi\xi}$$

$$u_{xy} = x^2 u_{\xi\eta}$$

$$\therefore u_{xx} + x^4 u_{yy} = x^4 (u_{yy} + u_{zz})_{\lambda} = 0$$

$$\therefore u_{yy} + u_{zz} = 0 - \frac{2u_y}{x^3}$$

10. Find the general solution of the partial differential equation

$$u_{xx} - u_{xy} = 0$$

$$R = 1, S = -\frac{1}{2}, T = 0, f = 0$$

$$\therefore S^2 - RT = \frac{1}{4} \therefore \text{hyperbolic}$$

$$\therefore \frac{dy}{dx} = \frac{-\frac{1}{2} \pm \sqrt{\frac{1}{4}}}{1} = 0 \text{ or } -1$$

$$\therefore \begin{cases} y = C_1 \\ y = -x + C_2 \end{cases} \quad \therefore \begin{cases} \xi(x, y) = y \\ \eta(x, y) = y + x \end{cases}$$

$$\therefore \xi_x = 0, \xi_y = 1, \eta_x = 1, \eta_y = 1$$

$$\therefore u_x = u_\xi \xi_x + u_\eta \eta_x = u_\eta$$

$$u_y = u_\xi \xi_y + u_\eta \eta_y = u_\xi + u_\eta$$

$$\therefore u_{xx} = u_{\eta\eta}$$

$$u_{xy} = u_{\eta\eta} + u_{\xi\eta}$$

$$\therefore u_{xx} - u_{xy} = -u_{\xi\eta} = 0 \quad \therefore u_{\xi\eta} = 0$$

$$\therefore u_\xi = \int 0 \, d\eta = F(\xi)$$

$$\therefore u = \int F(\xi) \, d\xi = f(\xi) + g(\eta) = f(y) + g(y+x)$$

11. Find the general solution to the partial differential equation

$$6u_{xx} - 5u_{xy} + u_{yy} = 1.$$

Now suppose that the equation satisfies the boundary conditions $u(x, 0) = \tanh(x)$ and $u_y(x, 0) = 3\operatorname{sech}^2(x)$. Find the particular solution.

$$R = 6, S = -\frac{5}{2}, T = 1, f = 1$$

$$\therefore S^2 - RT = \frac{25}{4} - 6 = \frac{1}{4} > 0 \quad \therefore \text{hyperbolic}$$

$$\therefore \frac{dy}{dx} = \frac{-S \pm \sqrt{S^2 - RT}}{R} = -\frac{1}{3} \text{ or } -\frac{1}{2}$$

$$\therefore \begin{cases} y = -\frac{1}{3}x + C_1 \\ y = -\frac{1}{2}x + C_2 \end{cases} \quad \therefore \begin{cases} \xi(x, y) = y + \frac{1}{3}x \\ \eta(x, y) = y + \frac{1}{2}x \end{cases}$$

$$\therefore \xi_x = \frac{1}{3}, \xi_y = 1, \eta_x = \frac{1}{2}, \eta_y = 1$$

$$\therefore u_x = u_\xi \xi_x + u_\eta \eta_x = \frac{1}{3}u_\xi + \frac{1}{2}u_\eta$$

$$u_y = u_\xi \xi_y + u_\eta \eta_y = u_\xi + u_\eta$$

$$\therefore u_{xx} = \frac{1}{9}u_{\xi\xi} + \frac{1}{3}u_{\eta\eta} + \frac{1}{3}u_{\xi\eta}$$

$$u_{yy} = u_{\xi\xi} + u_{\eta\eta} + 2u_{\xi\eta}$$

$$u_{xy} = \frac{1}{3}u_{\xi\xi} + \frac{1}{2}u_{\eta\eta} + \frac{5}{6}u_{\xi\eta}$$

$$\therefore 6u_{xx} - 5u_{xy} + u_{yy}$$

$$= \left(\frac{2}{3} - \frac{5}{3} + 1\right)u_{\xi\xi} + \left(\frac{3}{2} - \frac{5}{2} + 1\right)u_{\eta\eta} + \left(2 - \frac{25}{6} + 2\right)u_{\xi\eta}$$

$$= -\frac{1}{6}u_{\xi\eta} = 1$$

$$\therefore u_{\eta} = -6$$

$$\therefore u_{\xi} = \int -6 d\eta = -6\eta + F(\xi)$$

$$\therefore u = \int -6\eta + F(\xi) d\xi$$

$$= -6\xi\eta + f(\xi) + g(\eta)$$

$$= -6y^2 - 10xy - x^2 + f(y + \frac{1}{3}x) + g(y + \frac{1}{2}x)$$

$$\therefore u_y = -12y - 10x + f'(y + \frac{1}{3}x) + g'(y + \frac{1}{2}x)$$

$$\text{When } u(x, 0) = \tanh(x)$$

$$-x^2 + f(\frac{1}{3}x) + g(\frac{1}{2}x) = \tanh(x)$$

$$\therefore f(\frac{1}{3}t) + g(\frac{1}{2}t) = \tanh(t) + t^2$$

$$\therefore \frac{1}{3}f'(\frac{1}{3}t) + \frac{1}{2}g'(\frac{1}{2}t) = \text{sech}^2(t) + 2t$$

$$\text{When } u_y(x, 0) = 3\text{sech}^2(x)$$

$$\therefore f'(\frac{1}{3}t) + g'(\frac{1}{2}t) = 3\text{sech}^2(t) + 10t$$

$$\therefore \begin{cases} f'(\frac{1}{3}t) + \frac{3}{2}g'(\frac{1}{2}t) = 3\text{sech}^2(t) + 6t \\ f'(\frac{1}{3}t) + g'(\frac{1}{2}t) = 3\text{sech}^2(t) + 10t \end{cases}$$

$$\therefore \frac{1}{2}g'(\frac{1}{2}t) = -4t$$

$$\therefore g'(\frac{1}{2}t) = -8t \quad \therefore g'(h) = -16h$$

$$\therefore g(h) = -8h^2 + C$$

$$\therefore f'(\frac{1}{3}t) = 3\operatorname{sech}^2(t) + 18t$$

$$\therefore f'(h) = 3\operatorname{sech}^2(3h) + 54h$$

$$\therefore f(h) = \tanh(3h) + 27h^2 + C$$

$$\therefore u = -6y^2 - 10xy - x^2 + f(y + \frac{1}{3}x) + g(y + \frac{1}{2}x)$$

$$= -6y^2 - 10xy - x^2 + \tanh(y + \frac{1}{3}x) + 27(y + \frac{1}{3}x)^2$$

$$- 8(y + \frac{1}{3}x)^2$$